Class #9 - The Power Method

1. Assume A is real $n \times n$ and diagonalizable, so it has n linearly independent eigenvectors $\{\mathbf{v_1}, \cdots, \mathbf{v_n}\}$ in $\mathbb{R}^n$. Suppose its corresponding real eigenvalues are so that

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots.$$

Then, for any $\mathbf{x}$ in $\mathbf{R^n}$,

$$\mathbf{x} = c_1\mathbf{v_1} + \cdots + c_n\mathbf{v_n}$$

Assume $c_1 \neq 0$, then

$$A^k\mathbf{x} = A^k(c_1\mathbf{v_1} + \cdots + c_n\mathbf{v_n}) = c_1(A^k\mathbf{v_1}) + \cdots + c_n(A^k\mathbf{v_n})$$

$$= c_1(\lambda_1^k\mathbf{v_1}) + \cdots + c_n(\lambda_n^k\mathbf{v_n})$$

$$\frac{1}{\lambda_1^k}A^k\mathbf{x} = c_1\mathbf{v_1} + c_2\left(\frac{\lambda_2^k}{\lambda_1^k}\right)\mathbf{v_2} + \cdots + c_n\left(\frac{\lambda_n^k}{\lambda_1^k}\right)\mathbf{v_n}$$

Hence, for large k, the vector $A^k\mathbf{x}$ points almost in the same direction as $\mathbf{v_1}$ or $-\mathbf{v_1}$.

We can scale $A^k\mathbf{x}$ for each k so that the vector is not too long in its magnitude.

2. Power Method:

(Step 1) Choose $\mathbf{x_0}$.

(Step 2) For $k = 1, 2, 3, \cdots$ do until convergence

$$\hat{\mathbf{x}}_\mathbf{k} = A\mathbf{x_{k-1}}$$

$$\mathbf{x_k} = \hat{\mathbf{x}}_\mathbf{k}/\max(\hat{\mathbf{x}}_\mathbf{k})$$

(element that has max. abs. value)

3. Under the conditions: $c_1 \neq 0$, A diagonalizable, λ_1 is the only dominant eigenvalue, $\max(\hat{\mathbf{x}}_\mathbf{k}) \to \lambda_1$, and $\{\mathbf{x_k}\} \to \mathbf{w_1}$, a multiple of $\mathbf{v_1}$, as $k \to \infty$.

① Q # 4

② 5 min. break

③ Attendance

④ Par. quiz

⑤ Class # 9 handout ~

⑥ Q # 5 Next Mon. 10/19 (pp # 11 ~ 13)

⑦ Office hr Change : There are many seminars scheduled on Fri. afternoon.

Fri. 3-5 pm ⇒ Tues. 4:10 – 5:10
Wed. 11:10 – 12:10

(Thurs. 3-5 pm kept the same)

10/5

10/9

10/12: ① A is diagonalizable ? $A = PDP^{-1}$ Similar to $\begin{bmatrix} a & & \\ & \ddots & \\ & & b \end{bmatrix}$ e.v.

② e^{At} ⓐ Name: Matrix exponential

ⓑ Def : $I + At + \frac{(At)^2}{2!} + \cdots$

Class #7, #8 (pp)

Diff. Eqn

Since

$$\mathbf{x_k} = \frac{A^k x_0}{\max(A^k \mathbf{x_0})}, \quad \to \frac{\alpha \lambda_1 v_1}{?}$$

by the same discussion as in #1,

$$\mathbf{x_k} = \frac{A^k x_0}{\max(A^k \mathbf{x_0})} \to \alpha \mathbf{v_1}, \quad \text{and} \quad \max(\hat{\mathbf{x}}_\mathbf{k}) \to \lambda_1.$$

4. Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$, $\mathbf{x_0} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. eigenvalues: $0, -0.6235, 9.6235$. So, $\lambda_1 = 9.6235$, and $\mathbf{v_1} = \begin{bmatrix} 0.3851 \\ 0.5595 \\ 0.7339 \end{bmatrix}$.

$k = 1$: $\hat{\mathbf{x}}_\mathbf{1} =$ _____ $= \begin{bmatrix} 6 \\ 9 \\ 12 \end{bmatrix}$, $\max(\hat{\mathbf{x}}_\mathbf{1}) =$ _____, $\mathbf{x_1} =$ _____.

$k = 2$: $\hat{\mathbf{x}}_\mathbf{2} =$ _____ $= \begin{bmatrix} 5.00 \\ 7.25 \\ 9.50 \end{bmatrix}$, $\max(\hat{\mathbf{x}}_\mathbf{2}) =$ _____, $\mathbf{x_2} = \begin{bmatrix} 0.5263 \\ 0.7632 \\ 1.0000 \end{bmatrix}$.

$k = 3$: $\hat{\mathbf{x}}_\mathbf{3} =$ _____ $= \begin{bmatrix} 5.0526 \\ 7.3421 \\ 9.6316 \end{bmatrix}$, $\max(\hat{\mathbf{x}}_\mathbf{3}) =$ _____, $\mathbf{x_3} = \begin{bmatrix} 0.5246 \\ 0.7623 \\ 1.0000 \end{bmatrix}$.

$$\left| \max(\hat{x}_k) - \max(\hat{x}_{k-1}) \right| < 0.001$$

stop